Residual-limb FEA run report

knee_limb_4petal_T75

run id `knee_limb_4petal_T75` created 2026-10-25T20:31:13Z

Job state: CONVERGED. Solver: FEBio 4.13.0.32ae206ff. Load model: resolved_contact. **Numerical verification only: this report does not establish clinical accuracy, socket grip or tissue safety.** Geometry: supplied private limb model; layers are assumptions, not segmentation.

Warnings and labelled assumptions

- Layer thicknesses are measured from a supplied segmentation whose provenance and accuracy are not verified by this software.
- Socket geometry is a PROVISIONAL parametric model generated from configuration values; no verified current socket CAD was supplied (V6 excluded).
- No verified RevoFit rated strand tension supplied: rated-tension cases (25/50/75/100%) are NOT run.
- Strand tension 75 N is an EXAMPLE value, not a verified rating.
- Material 'skin' uses literature-based APPROXIMATIONS (not measured on this material): k.
- Material 'silicone_pad' uses literature-based APPROXIMATIONS (not measured on this material): c1, c2, k.
- Material 'distal_cushion' uses literature-based APPROXIMATIONS (not measured on this material): E, v.
- Each petal hinge is a frictionless revolute joint (penalty 1e4 N/mm, 1e7 N mm/rad). A radial stabiliser spring k = 0.05 N/mm per petal is NUMERICAL STABILISATION; its force and energy are reported.
- Cable span forces are applied to the rigid petals as fixed-direction resultant force and moment about each petal's centre of mass (directions do not follow petal rotation).

1 Inputs and provenance

input	path	sha256 (prefix)
config	/home/claude/knee-fea/configs/knee_limb_segmented.yaml	41856ea9e6eace08...
material_set	/home/claude/knee-fea/configs/materials/human_tissue.yaml	1d0b6fd1acd02755...
geometry:skin_envelope	/home/claude/knee-fea/data/limb/skin.stl	3c6c97941481aee8...
geometry:tibia_candidate	/home/claude/knee-fea/data/limb/part_1.stl	b226397ab5f9be13...
geometry:femur_candidate	/home/claude/knee-fea/data/limb/part_2.stl	d9733a28a9376598...
geometry:patella_candidate	/home/claude/knee-fea/data/limb/part_3.stl	865c6bbaa725fd09...
geometry:fibula_candidate	/home/claude/knee-fea/data/limb/part_4.stl	4e0e4298aaea7958...

Repository commit: `ddacd182835d4ed535d9bd446d6d31699754ccfa` (dirty worktree: True). Solver input `solver/model.feb` sha256 `f842ae782f3574a28463....` Execution command: `/usr/local/bin/febio4 -nosplash -silent -noconfig -i model.feb -dump model.dmp`.

1.1 Geometry audit

body	status	source unit	dims mm	vol mm3	closed	comp.	s
skin	VALID_CLOSED_SURFACE	mm	143.7x152.1x289.1	3.14968e+06	True	1	0
tibia_candidate	DEFECTIVE	mm	88.09x62.48x150	274532	False	7362	6
femur_candidate	DEFECTIVE	mm	95.21x74.1x114.8	300022	False	7786	6

body	status	source unit	dims mm	vol mm3	closed	comp.	s
patella_candidate	DEFECTIVE	mm	47.69x23.71x47.47	25492	False	2246	1
fibula_candidate	DEFECTIVE	mm	28.26x26.2x100.4	19938.1	False	2448	2
skin_inner	VALID_CLOSED_SURFACE	mm	136.5x144.5x274.6	2.70046e+06	True	1	N
fat_inner	VALID_CLOSED_SURFACE	mm	129.7x137.3x260.9	2.00289e+06	True	1	N

Anatomy roles (as configured): tibia_candidate: tibia (HEURISTIC label, unconfirmed; 150 mm long, cavity in muscle.stl), femur_candidate: distal femur (HEURISTIC label, unconfirmed), patella_candidate: patella (HEURISTIC label, unconfirmed; cavity in muscle.stl), fibula_candidate: fibula (HEURISTIC label, unconfirmed; slender, posterolateral to part_1).

body	fraction inside skin	min clearance mm	skin intersections
tibia_candidate	1	6.26	0
femur_candidate	1	1.474	0
patella_candidate	1	3.947	0
fibula_candidate	1	8.126	0

2 Model

Layer thicknesses are MEASURED per skin vertex from the supplied segmented surfaces (skin inner and fat inner); the segmentation itself is unverified. Skin thickness {"min": 0.674, "p01": 1.344, "median": 3.746, "max": 9.708} mm; fat thickness {"min": 0.15, "p01": 0.15, "median": 3.08, "max": 11.572} mm. 50 skin vertices had layer thickness reduced near bone. Minimum wedge scaled Jacobian 0.03013.

id	block	element	material	region	formulation
0	skin	penta6	skin	skin	three-field-solid
1	fat	penta6	fat	fat	three-field-solid
2	proximal_cap	penta6	muscle	muscle (layers under artificial proximal cut)	three-field-solid
3	muscle	tet4	muscle	muscle	ut4-solid
4	bone_tibia_candidate	tet4	rigid_bone_tibia_candidate	bone:tibia_candidate	bone
5	bone_femur_candidate	tet4	rigid_bone_femur_candidate	bone:femur_candidate	bone
6	bone_patella_candidate	tet4	rigid_bone_patella_candidate	bone:patella_candidate	bone
7	bone_fibula_candidate	tet4	rigid_bone_fibula_candidate	bone:fibula_candidate	bone
8	petal1_pad	hex8	silicone_pad	socket:petal1:silicone pad	three-field-solid
9	petal1_shell	hex8	rigid_petal1	socket:petal1:rigid petal	rigid
10	petal2_pad	hex8	silicone_pad	socket:petal2:silicone pad	three-field-solid
11	petal2_shell	hex8	rigid_petal2	socket:petal2:rigid petal	rigid
12	petal3_pad	hex8	silicone_pad	socket:petal3:silicone pad	three-field-solid
13	petal3_shell	hex8	rigid_petal3	socket:petal3:rigid petal	rigid
14	petal4_pad	hex8	silicone_pad	socket:petal4:silicone pad	three-field-solid
15	petal4_shell	hex8	rigid_petal4	socket:petal4:rigid petal	rigid
16	distal_cushion	penta6	distal_cushion	socket:distal cushion	three-field
17	distal_cup	penta6	rigid_cup	socket:rigid distal cup (+ puck/plate/adapter as support)	rigid

Strain-energy functions as implemented by FEBio 4.13 (checked against source and against an independent implementation for homogeneous deformations, Section ??). Isochoric invariants use $\tilde{\mathbf{C}} = J^{-2/3}\mathbf{C}$.

$$\text{HGO (skin):}\quad W = \tfrac{c}{2}(\tilde{I}_1 - 3) + \frac{k_1}{2k_2}\sum_{a=1,2}\left[e^{k_2\langle E_a\rangle^2} - 1\right] + \tfrac{k}{2}\left[\tfrac{J^2-1}{2} - \ln J\right], \quad E_a = \kappa(\tilde{I}_1 - 3) + (1 - 3\kappa)$$

$$\text{Ogden (fat, muscle):}\quad W = \sum_i \frac{c_i}{m_i^2}\left(\tilde{\lambda}_1^{m_i} + \tilde{\lambda}_2^{m_i} + \tilde{\lambda}_3^{m_i} - 3\right) + \tfrac{k}{2}(\ln J)^2$$

$$\text{Mooney–Rivlin (silicone, cushion):}\quad W = c_1(\tilde{I}_1 - 3) + c_2(\tilde{I}_2 - 3) + \tfrac{k}{2}(\ln J)^2$$

Fibre families lie at $\pm\gamma$ from local axis $\mathbf{e}_1$ in the $\mathbf{e}_1$–$\mathbf{e}_2$ plane and resist only tension ($E_a \geq 0$). FEBio's c equals ABAQUS $2C_{10}$; FEBio's HGO volumetric term equals ABAQUS GOH with $D = 2/k$. The sources' conventions and the conversions applied are listed per parameter in Table 2.

material	FEBio model	param	value (MPa/deg/-)	source	note
skin	Holzapfel-Gasser-Ogden	c	0.2014	NiAnnaidh2012	source mu; FEBio ground matrix V - 3), so c = mu (Abaqus C10 = m
skin	Holzapfel-Gasser-Ogden	k1	24.53	NiAnnaidh2012	source reports k1 = 243.6 in its c isation (anisotropic term scaled gives the ABAQUS value 24.53 ABAQUS value is used. FE
skin	Holzapfel-Gasser-Ogden	k2	0.1327	NiAnnaidh2012	
skin	Holzapfel-Gasser-Ogden	kappa	0.1404	NiAnnaidh2012	dispersion, 0 ¡= kappa ¡= 1/3
skin	Holzapfel-Gasser-Ogden	gamma	41	NiAnnaidh2012	angle of each fibre family from l (FEBio gamma). Mapping of Lar the lower leg to e1 is an ASSUM fiber_reference_axis
skin	Holzapfel-Gasser-Ogden	k	20.14	TissueBulkRatioApprox	bulk modulus = 100 c, the huma ratio used for fat and muscle (appr FEBio HGO uses U = k/2((J^2- (= ABAQUS GOH with
fat	Ogden	c1	0.00522	Sengeh2016	source Psi = (c/m^2) sum(l^^m 2) == FEBio Ogden with c1=c2 m2=-m; small-strain shear modul
fat	Ogden	m1	4.79	Sengeh2016	
fat	Ogden	c2	0.00522	Sengeh2016	
fat	Ogden	m2	-4.79	Sengeh2016	
fat	Ogden	k	0.522	Sengeh2016	source constrained bulk = 100 c w (ln J)^2, identical to FEBio press
muscle	Ogden	c1	0.0052	Sengeh2016	
muscle	Ogden	m1	4.78	Sengeh2016	
muscle	Ogden	c2	0.0052	Sengeh2016	
muscle	Ogden	m2	-4.78	Sengeh2016	
muscle	Ogden	k	0.52	Sengeh2016	
bone_deformable	neo-Hookean	E	1.7e+04	ReillyBurstein1975	human cortical bone, longitudinal the WHOLE bone volume (cortica not distinguished) - simplification
bone_deformable	neo-Hookean	v	0.3	ReillyBurstein1975	
silicone_pad	Mooney-Rivlin	c1	0.01165	Sanders2004	neo-Hookean (c2 = 0): FEBio Mooney-Rivlin small-strain shear = 2(c1 + c2); mu = 23.3 kPa = m shear stiffness G of the
silicone_pad	Mooney-Rivlin	c2	0	Sanders2004	neo-Hookean choice: the source re fits only
silicone_pad	Mooney-Rivlin	k	2.33	TissueBulkRatioApprox	silicone gels are near-incompressib mu by the same convention as the
distal_cushion	neo-Hookean	E	0.071	GyroidHomogenisationApprox	E* = Es (rho*)^2 with Es = 6 (11.1 MPa (small-strain Young's the NinjaFlex 85A fit, c1 = 0.385 MPa, Reppel & Weinberg
distal_cushion	neo-Hookean	v	0.3	GyroidHomogenisationApprox	a sparse lattice is compressible near-incompressible like solid TP foam Poisson ratio

References: [NiAnnaidh2012] Ni Annaidh A, Bruyere K, Destrade M, Gilchrist MD, Maurini C, Ottenio M, Saccomandi G. Automated estimation of collagen fibre dispersion in the dermis and its contribution to the anisotropic behaviour of skin. Ann Biomed Eng. 2012;40(8):1666-78. doi:10.1007/s10439-012-0542-3. [Sengeh2016] Sengeh DM, Moerman KM, Petron A, Herr H. Multi-material 3-D viscoelastic model of a transtibial residuum from in-vivo indentation and MRI data. J Mech Behav Biomed Mater. 2016;59:379-392. doi:10.1016/j.jmbbm.2016.02.020. [ReillyBurstein1975] Reilly DT, Burstein AH. The elastic and ultimate properties of compact bone tissue. J Biomech. 1975;8(6):393-405. doi:10.1016/0021-9290(75)90075-5. [Sanders2004] Sanders JE, Nicholson BS, Zachariah SG, Cassisi DV, Karchin A, Fergason JR. Testing of elastomeric liners used in limb prosthetics: classification of 15 products by mechanical performance. J Rehabil Res Dev. 2004;41(2):175-186.. [ReppelWeinberg2018] Reppel T, Weinberg K. Experimental determination of elastic and rupture properties of printed Ninjaflex. Technische Mechanik. 2018;38(1):104-112. doi:10.24352/UB.OVGU-2018-010. [GyroidHomogenisationApprox] Approxima-

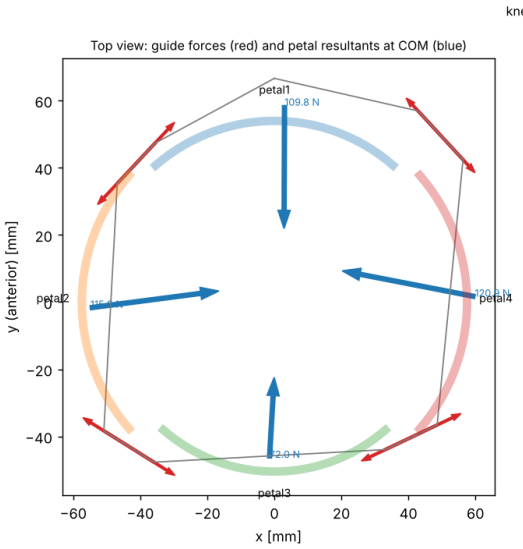
tion: Gibson-Ashby cellular-solid scaling E*/Es = C (rho*)^n with C = 1 and n = 2 (bending-dominated lattice) applied to 8% gyroid infill of 85A TPU (Es from Reppel & Weinberg 2018). Range n = 1.5 (stretch/sheet-like) to C = 0.5 gives 36-252 kPa. Gibson LJ, Ashby MF. Cellular Solids: Structure and Properties, 2nd ed., Cambridge University Press, 1997. doi:10.1017/CBO9781139878326. [TissueBulkRatioApprox] Approximation: human soft-tissue bulk modulus taken as 100 x shear modulus, the same near-incompressibility ratio Sengeh et al. 2016 used for fat and muscle. doi:10.1016/j.jmbbm.2016.02.020. [PriorPackageAssumption] Supplied handoff package 03_Previous_Tissue_Analysis (silicone mu = 100 kPa listed as an assumed material; no measurement)..

2.1 Socket (provisional parametric model)

PROVISIONAL parametric socket (not verified design geometry).

3 Loading

Applied strand tension 75.0 N (basis: example). Rated tension: None. Output levels: [18.75, 37.5, 56.25, 75.0] N. PROVISIONAL default single-row lace generated from the parametric petals



Lace tension 75.0 N (example); guide friction mu_g = 0.0

body	Fx N	Fy N	Fz N	Mx Nmm	My Nmm	Mz Nmm
petal1	-0.44	-109.77	0	1178.3	-4.7	-575.2
petal2	114.92	14.91	0	-203.7	1569.6	709.5
petal3	4.19	71.88	0	-1881.8	109.7	-510.7
petal4	-118.67	22.98	0	-285.9	-1476.8	-350.8
SUM F	0.00e+00	-7.11e-15	0.00e+00			
SUM M@O				-9.09e-13	0.00e+00	0.00e+00

Figure 1: Lace route, guide forces and petal resultants (free-body check: sums must vanish).

4 Solver and convergence

Tolerances: dtol=0.01, etol=0.01, rtol=0.001; Newton with auto (max_ups=0); PARDISO; automatic time stepping with must points; max retries 10. Retries in log: 0; negative-Jacobian events: 0.

step	time	converged	iterations	final residual	final energy norm
1	0.02	yes	7	0.0002486	2.02e-05
2	0.05	yes	5	0.0004679	1.597e-05
3	0.1	yes	6	0.002534	3.609e-05
4	0.15	yes	4	0.0001426	2.936e-05
5	0.2	yes	4	2.038e-05	7.954e-06
6	0.25	yes	4	0.00126	0.000275
7	0.3	yes	5	0.008181	0.0004522
8	0.35	yes	5	0.04365	0.0003975
9	0.4	yes	4	2.599e-06	6.615e-07
10	0.45	yes	4	8.876e-07	1.631e-07
11	0.5	yes	4	1.096e-06	2.727e-07
12	0.55	yes	4	5.907e-07	6.288e-08
13	0.6	yes	6	1.055	0.0009087
14	0.65	yes	6	0.02007	0.001533
15	0.7	yes	3	0.0003777	9.924e-05

step	time	converged	iterations	final residual	final energy norm
16	0.75	yes	4	0.02173	0.003564
17	0.8	yes	5	0.1714	0.001358
18	0.85	yes	5	1.057	0.001584
19	0.9	yes	5	0.009493	0.0006443
20	0.95	yes	4	0.02688	0.0008536
21	1	yes	4	0.0005463	0.0003979
1	1.02	yes	11	0.09645	8.601e-05
2	1.039	yes	5	0.00592	0.0005135
3	1.1	yes	5	2.279e-09	4.345e-10
4	1.2	yes	4	4.78e-06	3.625e-07
5	1.3	yes	3	0.0003277	9.271e-05
6	1.4	yes	4	6.697e-06	7.768e-07
7	1.5	yes	4	3.815e-06	1.443e-07
8	1.6	yes	4	3.536e-08	1.442e-08
9	1.7	yes	5	0.02973	0.001189
10	1.8	yes	4	1.109e-06	1.909e-06
11	1.9	yes	5	1.531	0.005222
12	2	yes	4	1.521e-05	2.721e-06

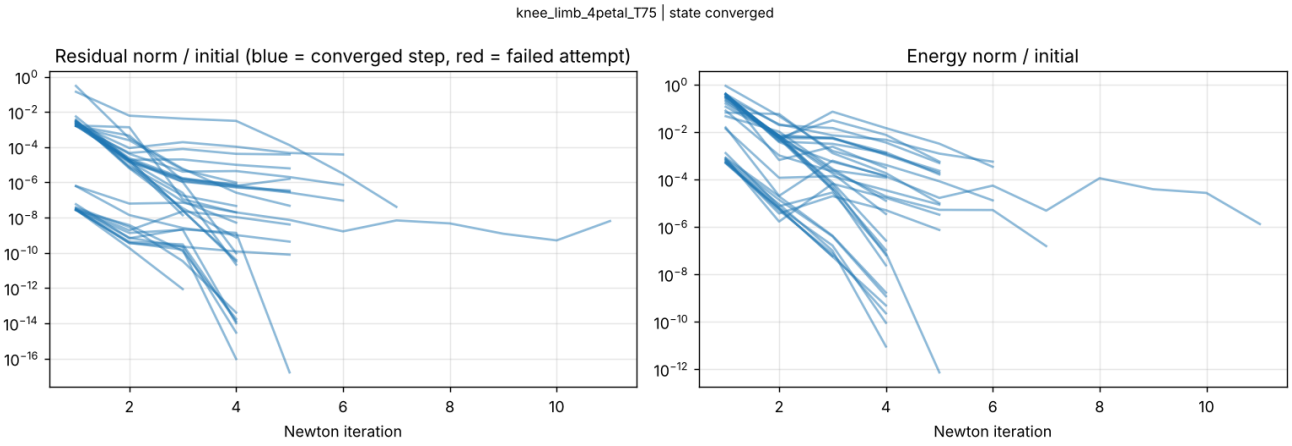


Figure 2: Newton convergence history.

5 Results (actual FEBio fields)

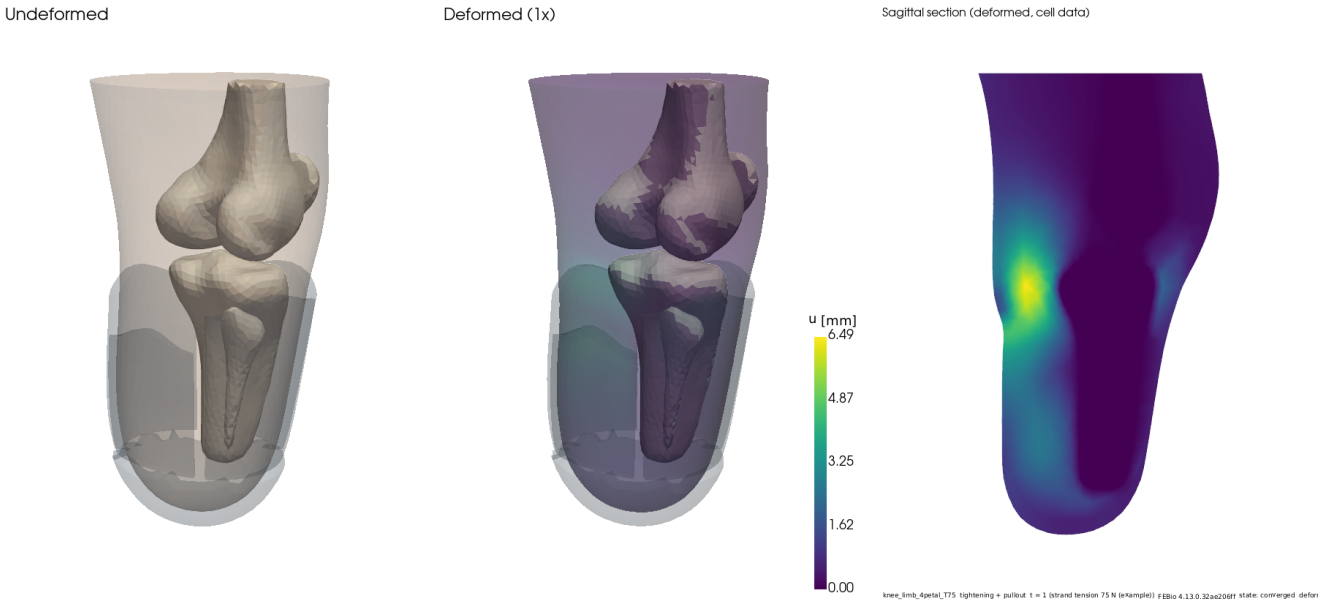


Figure 3: Full limb, undeformed and deformed at 1× (true scale), with sagittal section.

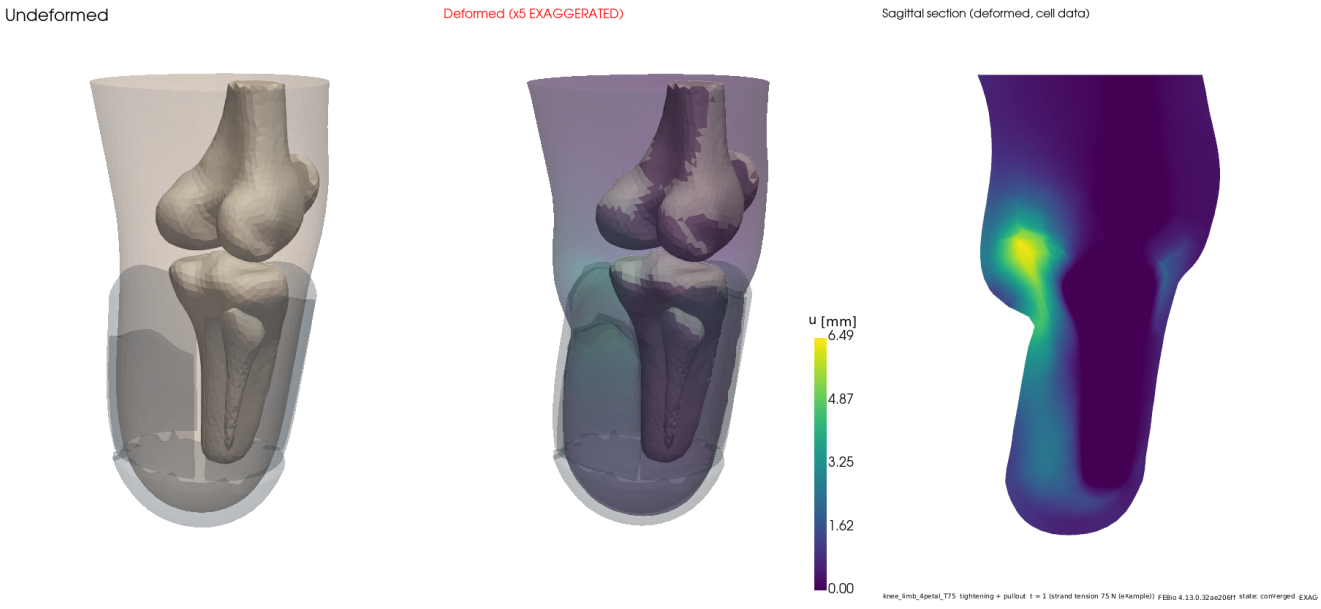


Figure 4: **EXAGGERATED** deformation (labelled scale) for visual inspection only.

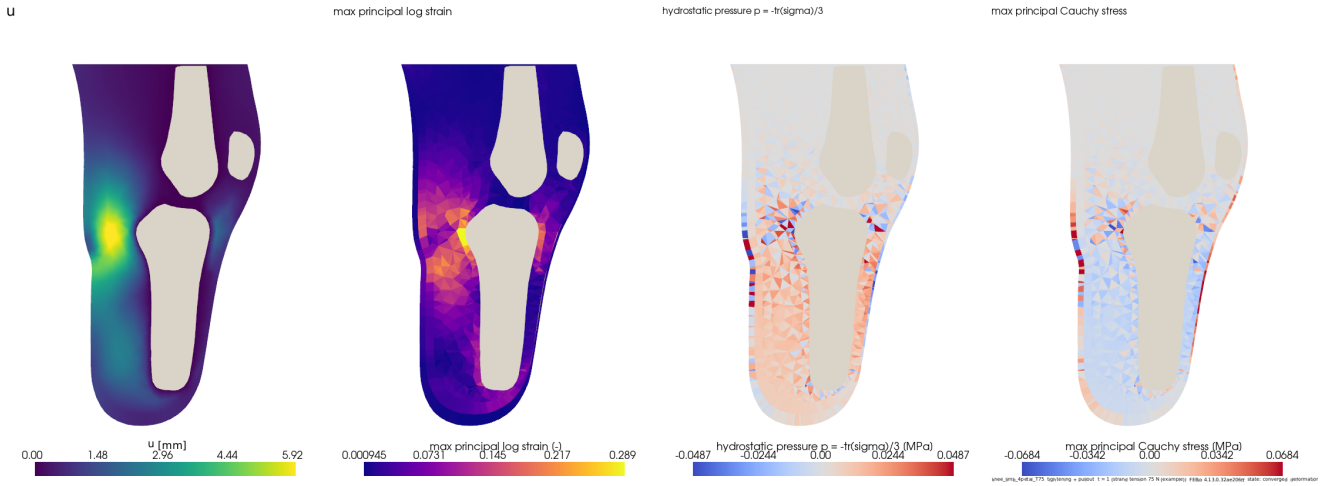


Figure 5: Sections through the deformed limb: displacement, maximum principal log strain, hydrostatic pressure, maximum principal Cauchy stress (cell data, no smoothing). Peak values may be mesh-dependent and are not injury predictions.

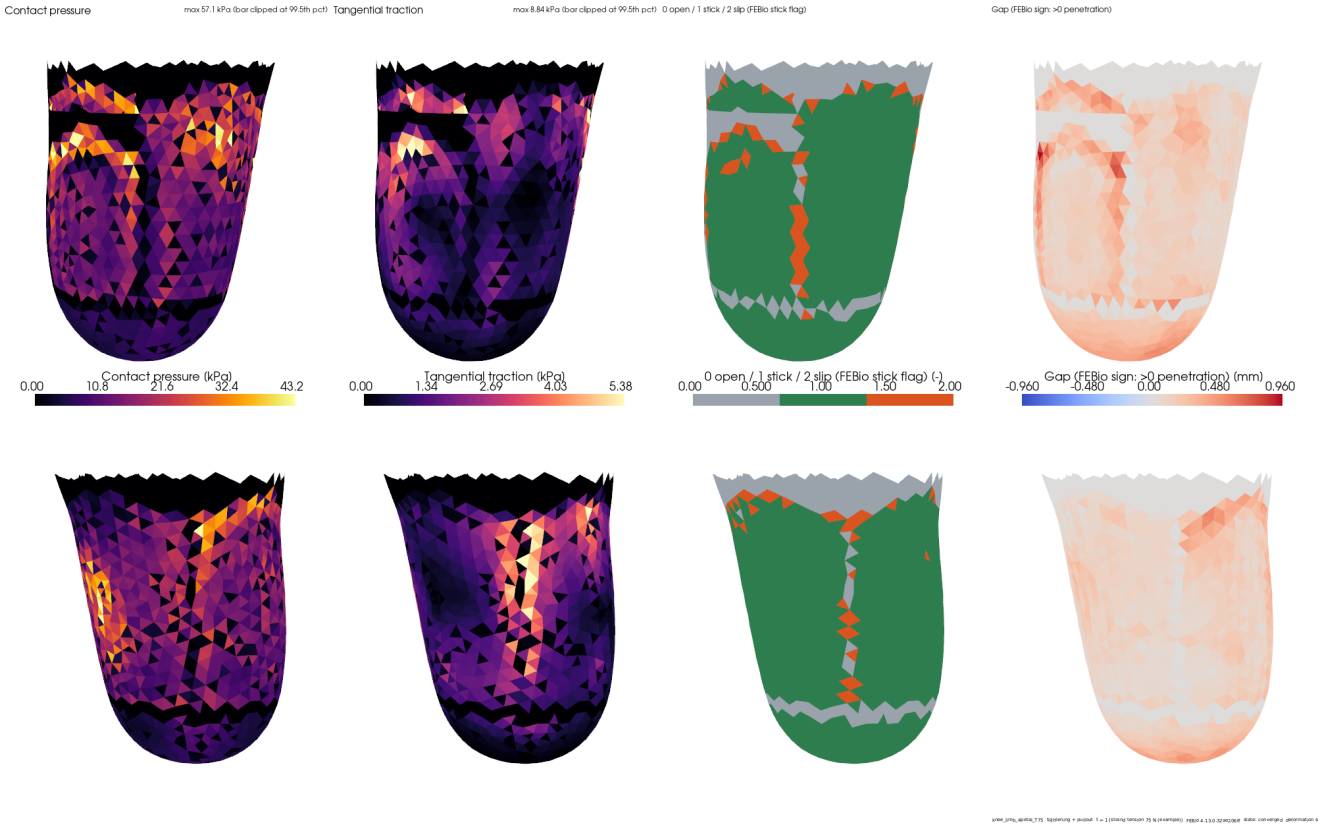


Figure 6: Skin contact pressure, tangential traction, derived stick/slip status and gap.

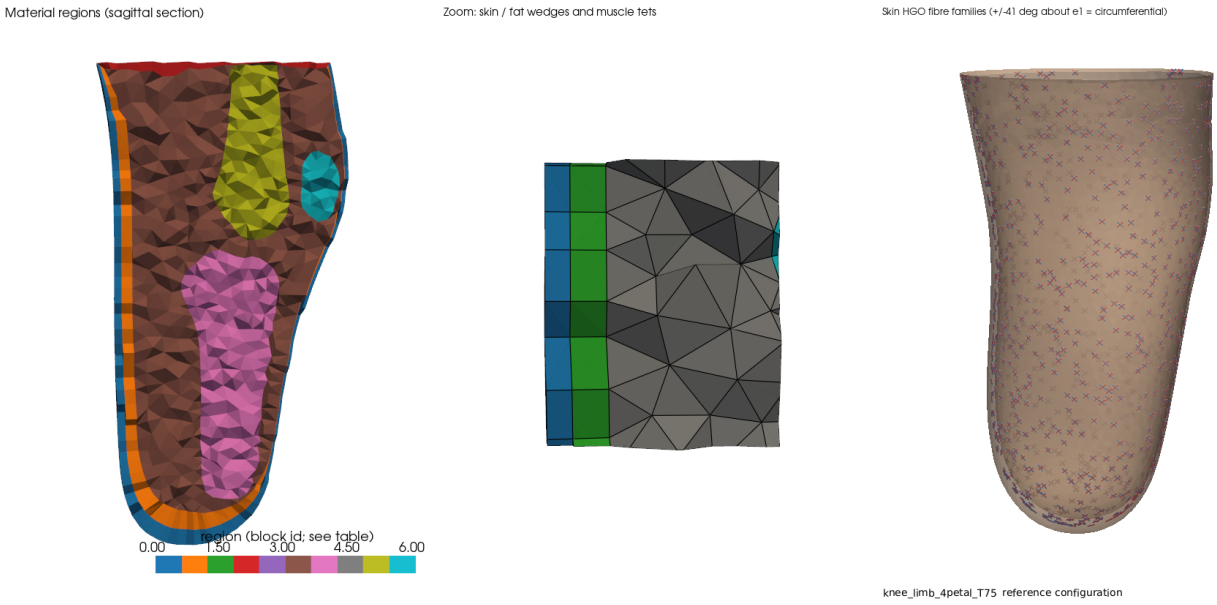


Figure 7: Material regions and skin fibre orientation.

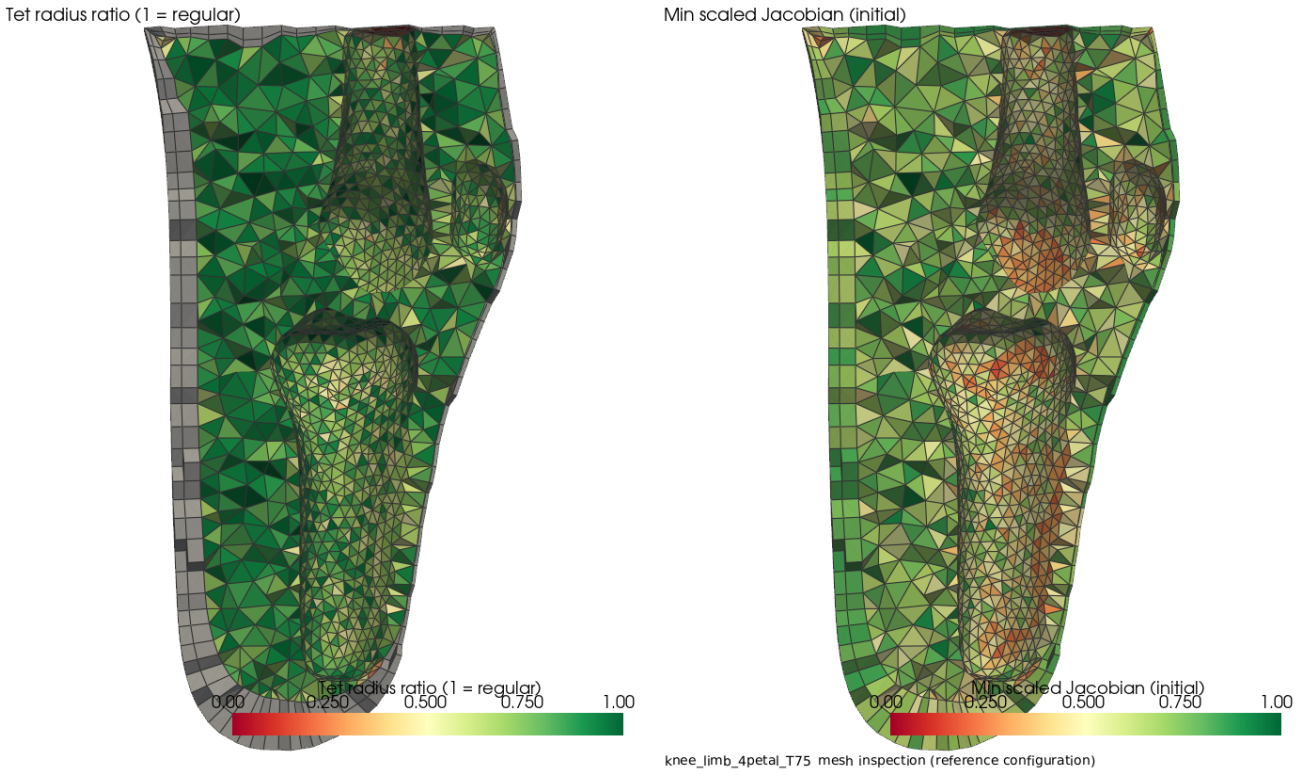


Figure 8: Mesh element quality.

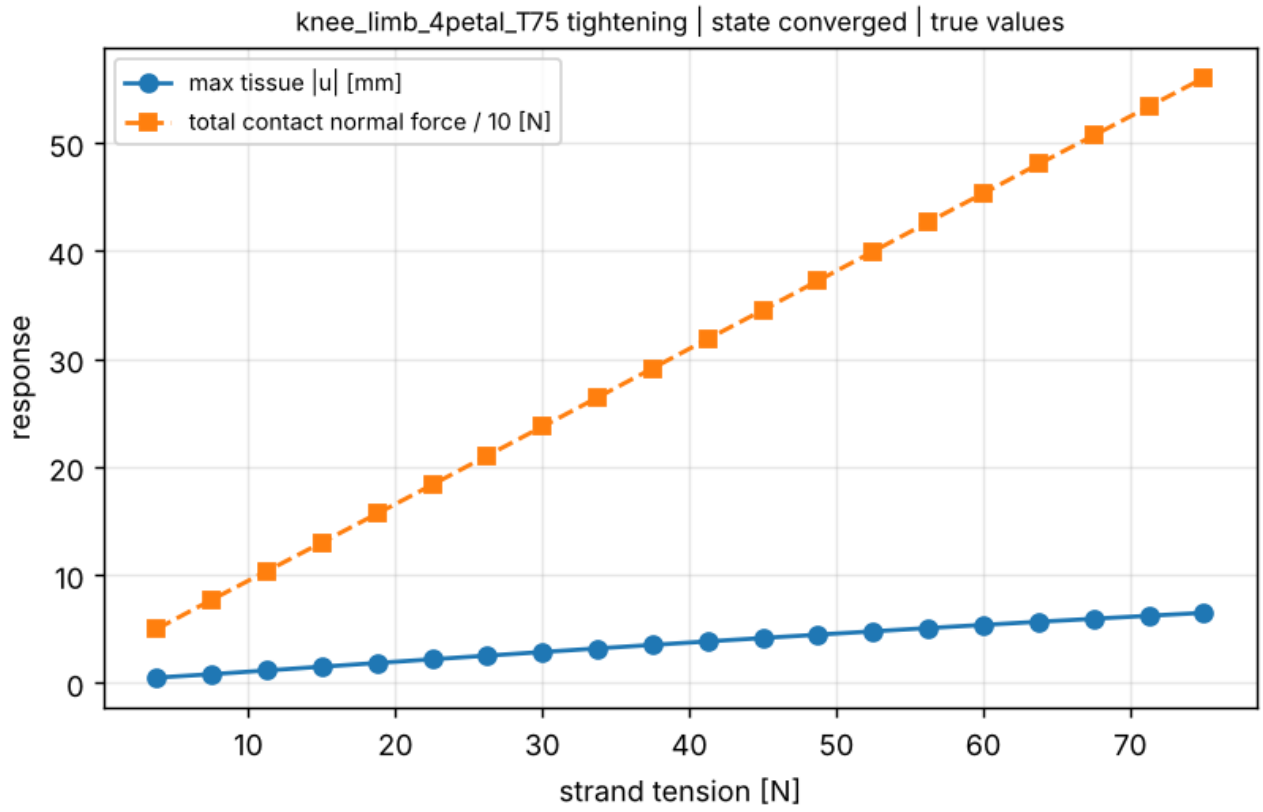


Figure 9: Force–displacement / load–response curves.

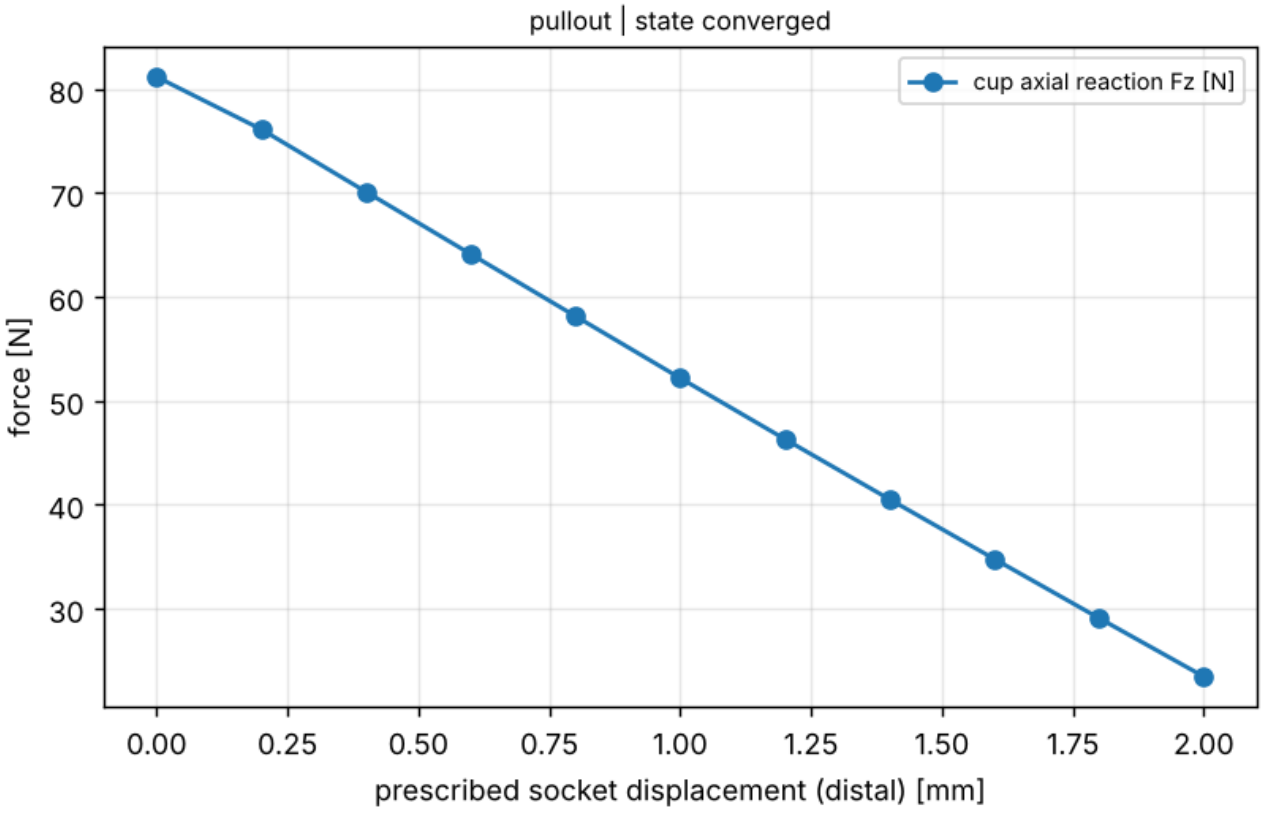


Figure 10: Pull-out: reaction versus relative socket displacement.

t	max —u— tissue mm	min J	inverted	contact N	max p kPa	slip area frac.	force imbalance
0.05	0.4909	0.9696	0	49.89	13.71	0.3534	1.375e-06
0.1	0.8061	0.9605	0	77	16.19	0.2979	1.383e-07
0.15	1.153	0.9417	0	103.5	17.67	0.2566	2.959e-06
0.2	1.5	0.9233	0	130.1	20.87	0.2146	6.585e-08
0.25	1.844	0.9054	0	156.7	24.12	0.1849	7.833e-06
0.3	2.185	0.888	0	183.5	27.1	0.1571	4.808e-06
0.35	2.525	0.8707	0	210.5	30.12	0.1459	7.875e-07
0.4	2.86	0.8538	0	237.3	31.51	0.1237	8.418e-07
0.45	3.192	0.8376	0	264.3	34	0.1115	1.144e-06
0.5	3.518	0.8203	0	291.4	36.5	0.09919	1.643e-10
0.55	3.841	0.8022	0	318.5	38.89	0.09096	6.542e-10
0.6	4.156	0.7856	0	345.3	41.38	0.08522	2.639e-05
0.65	4.467	0.7697	0	372.4	43.95	0.08068	3.731e-07
0.7	4.774	0.7529	0	399.5	46.51	0.07657	1.223e-07
0.75	5.077	0.7366	0	426.7	49.08	0.07453	1.881e-06
0.8	5.372	0.7207	0	453.4	51.67	0.06783	3.314e-06
0.85	5.661	0.7053	0	480.5	54.22	0.06381	1.414e-05
0.9	5.938	0.6903	0	506.8	55.48	0.0609	2.39e-06
0.95	6.216	0.6761	0	533.6	55.45	0.0588	3.212e-06
1	6.49	0.6635	0	560.3	57.11	0.05726	6.97e-07
1.1	6.427	0.6585	0	558.9	56.83	0.05405	4.281e-11
1.2	6.297	0.6556	0	554.1	56.68	0.05503	5.631e-10
1.3	6.19	0.6531	0	549	57.2	0.05418	2.615e-05
1.4	6.083	0.6503	0	543.8	57.72	0.05978	1.178e-06
1.5	5.985	0.6479	0	538.5	58.42	0.06401	1.562e-06
1.6	5.875	0.6455	0	533	58.89	0.06627	1.79e-09
1.7	5.771	0.6431	0	527.4	58.57	0.06754	9.588e-07
1.8	5.669	0.6409	0	521.9	58.87	0.07073	8.792e-10
1.9	5.562	0.6387	0	516.3	59.51	0.07457	1.078e-05
2	5.461	0.6365	0	510.5	55.2	0.08054	1.913e-09

Petal free bodies at t = 1 (force magnitudes, N). Pad contact force is the facet-averaged traction on the skin zone times the deformed facet area (an approximate post-processed integral); the global reaction balance in the table above is the exact check.

petal	—cable—	—hinge—	—pad contact—	—stabiliser—	—residual—	residual/cable
petal1	110	32.7	129	0.0415	0.000869	7.92e-06
petal2	116	17.5	119	0.177	0.000197	1.7e-06
petal3	72	25.1	93.9	0.211	0.00226	3.14e-05
petal4	121	13.3	121	0.0774	0.00249	2.06e-05

block	max —u—	max log e	min log e	max p MPa	min p MPa	max vM MPa	min J
skin	5.126	0.0757	-0.08482	0.08836	-0.1546	0.1626	0.9957
fat	5.101	0.4986	-0.5342	0.05752	-0.006823	0.02474	0.9061
proximal_cap	1.153	0.06398	-0.04491	0.00546	-0.01205	0.0009935	0.9913
muscle	5.461	0.6037	-0.8818	0.369	-0.1187	0.08425	0.6365
petal1_pad	2.507	0.1455	-0.171	0.03268	-1.981e-05	0.01303	0.9863
petal2_pad	5.245	0.1582	-0.2686	0.04344	0.002593	0.01947	0.9813
petal3_pad	6.165	0.1557	-0.2049	0.0509	0.001841	0.01774	0.9819
petal4_pad	3.354	0.1537	-0.2202	0.03515	0.001148	0.01591	0.9854
distal_cushion	2.425	0.02096	-0.08076	0.003824	-2.906e-05	0.004951	0.9393

6 Limitations

- Numerical verification is separate from experimental or clinical validation; no validation data were supplied.

- Tissue layers are geometric assumptions; material parameters come from literature on other subjects/sites (see table) or are labelled assumptions.
- The socket is a provisional parametric model; cable routing is provisional unless the configuration supplies verified routing.
- Quasi-static, no viscoelasticity, no perfusion, no active muscle, no skin adhesion or liner; friction is Coulomb with a screening coefficient.
- A converged state does not establish grip; pull-out is evaluated only through the resolved contact tractions and relative motion of this model.